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(Printed Pages 2)

(20517)

Roll No. 9791016

B.Sc. (Micro.)-III Year

3510

B.Sc. (Micro.) Examination, May 2017

Microbial Ecology

(B-307)

35

Time : Three Hours]

[Maximum Marks : 50

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. What is air microbiology? Discuss different sources of airborne microbes. 10

2. Write short notes on : 5+5

(i) Alimentary tract microflora

(ii) Phosphorus cycle

3. How does the development of microbial communities take place in the environment?

10

4. Define the term 'sewage'. Give its chemical composition and methods of treatment. 6

10

5. Write short notes on : 5+5
7

(i) BOD and COD

(ii) Microbe-plant interaction and its uses

6. Enumerate the microbial population of the rumen of animals. 10

7. Discuss the succession of microflora on decomposing matter. 10

8. Explain the role of biogeochemical cycles with reference to nitrogen cycle. 8
10

9. Write short notes on : 5+5
7

(i) Waterborne pathogenic microbes

(ii) Types of biological interactions

10. Describe the microbes found in sewage. 10

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B.Sc. (Microbiology) IIIrd Year Examination,
May-2019

MICROBIAL ECOLOGY

(B-307)

[B.Sc. (Micro.)]

Time : 3 Hrs.]

[M.M. : 50

Note :- Attempt any *five* questions. All questions carry equal marks.

1. Describe the type of micro-organisms in air/ atmosphere.
2. Describe the microbial population found in alimentary tract.
3. Describe major microbes found in Rumen.

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4. Describe in detail Microbe-Microbe interactions.

Support your answer with suitable examples.

5. Write a note on Microbe-animal interactions.

6. What is the role of micro-organisms in cycling of carbon ? Support your answer with suitable examples.

7. What is the role of micro-organisms in cycling of Sulphur and Iron ?

8. Describe the various compositions of sewage.

9. Describe in detail the two fungi involved in sewage.

10. Write a note on COD and BOD of sewage.

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Roll No.

B.Sc.(Micro.)-III Year

3510 (CV)

B.Sc. (Microbiology) IIIrd Year

Examination, June- 2020

MICROBIAL ECOLOGY

(B-307)

[B.Sc. (Micro.)]

Time : Two Hours]

[Maximum Marks : 50

Note : Attempt any **four** questions. **All** questions carry equal marks.

1. Discuss the role of micro-organisms in environment.
2. What is bioremediation? How can it be useful for us.
3. Write a note on microbial diversity in the digestive tract of herbivores.
4. Write a note on the role of microorganisms in recycling of elements.

P.T.O.

5. Discuss how sewage is treated. What are primary, secondary and tertiary treatment.
6. Describe the contribution of each partners in the a) lichen and b) mycorrhizae symbiosis systems.
7. Write a note on the different types of microorganisms in sewage.
8. What is BOD and COD? How does it affect the pollution problem.
9. Write a note on microbe-microbe interactions.
10. Write a note on the cycling of carbon, nitrogen and iron.

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(Printed Pages 3)

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Roll No.

B.Sc.(Micro.)-III Yr.

3510(CV-II)

B.Sc. (Microbiology) Examination,

June - 2021

MICROBIAL ECOLOGY

(B-307)

[B.Sc. (Micro.)]

Time : 1½ Hours]

[Maximum Marks : 50

Note : Attempt any **two** questions. **All**
questions carry equal marks.

1. Describe the various aquatic micro-organisms.
2. What key roles do microorganisms play in soil ecosystems.

P.T.O.

3. Write a note on:

(a) Microbial-retebrae interaction

OR

(b) Microbial-Plant interaction

4. Write a note on role of microorganism in nitrogen cycle.

5. Discuss the various processes adapted to:

(a) Primary treatment of water

(b) Secondary treatment of water

(c) Tertiary treatment of water

6. Write a note on role of microorganisms in biodegradation enhancement.

7. Write a note on microorganisms in terrestrial ecosystem.

8. Write a note on the microorganism in sewage.

9. What is the role of microorganisms in cycling of carbon, phosphorus and iron.

10. What do you understand by COD and BOD of sewage. How does it affect the process of sewage treatment.

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**B.Sc. (Microbiology) Examination,
June-2022**

Microbial Ecology

(B-307)

[B.Sc. (Micro.)]

Time : Three Hours]

[Maximum Marks : 50

Note : Attempt any **five** questions. **All**
questions carry equal marks.

1. What are bacteria? Describe the types of bacteria which are inhabitant in rumen of cattle with suitable example.

P.T.O.

2. List the name of important groups of microorganisms which are commonly found in terrestrial environment.
3. What are decomposers and why is it important to ecosystem? Explain with suitable example.
4. Define the term Rhizobium? What is the importance of Rhizobium in ecosystem?
5. Describe microbes-microbes interaction with suitable example.
6. Outline the iron cycle, and explain the roles of microorganisms in this cycle.
7. Define ammonification, nitrification, denitrification and nitrogen fixation.

8. What is sludge? Describe the activated sludge processes for sewage treatment?
9. Define the term bioremediation? Name several important microbial groups and their role in sewage treatment system?
10. What are COD and BOD and how are these similar and different?

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**B.Sc. (Microbiology)
Examination, May-2024
MICROBIAL ECOLOGY
(B-307)**

[B.Sc. (Micro.)]

Time : Three Hours]

[Maximum Marks : 50

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. Describe the microbial growth patterns in aquatic environment. 10
2. Discuss secondary treatment of Municipal waste. 10
3. Describe in detail about Ruman microbiology. 10

P.T.O.

4. What is biogeochemical cycle? Discuss the role of micro organism in Nitrogen Cycle. 10
5. Describe in detail the two bacteria involved in Sewage. 10
6. Describe in detail Plant Microbe interactions support your answer with suitable examples. 10
7. What is microbe animal interaction describe the location of normal microbiota on and in human body? 10
8. Describe the microbial population found in alimentary tract. 10
9. Describe the microorganism encountered in terrestrial environment. What special adaptations microorganism have developed to withstand scarcity of water. 10

10. Write a note on COD and BOD of
Sewage.

10

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Anuj

B.Sc. (Microbiology)
Examination, April-2025

MICROBIAL ECOLOGY
(B-307)

[B.Sc. (Micro.)]

Time : Three Hours]

[Maximum Marks : 50

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. Define Microbial Interactions & describe each type with the help of examples. 10
2. What are the ecological implication of microbial antibiotic resistance in natural environments. 10

P.T.O.

3. Define nutrient cycling & its importance.
Explain Nitrogen cycle in detail. 10
4. List various general & specific factors
that will influence the microbial ecology
of following : 10
- (a) Terrestrial environment
 - (b) Aquatic environment
 - (c) Air
5. Explain how microbial symbiosis e.g.
Mycorrhiza, Rhizobia etc. affect nutrient
cycling in natural environments. 10
6. Define the impact of microbes on soil
fertility & plant productivity. 10
7. What are the different kinds of sewage
systems. Explain with the help of
examples. 10
8. Differentiate b/w COD and BOD. & list
various microbes in sewage. 10

9. Explain in detail various types of Microbe-plant interaction with the help of examples. 10
10. In what ways do microbes contribute to the stability & resilience of ecosystem facing environmental stress (e.g. pollution, climate change). 10